

Section: **3.1 – STRENGTH OF METALLIC PARTS**

Chapter: **3110**

Title: **BEAMS SECTIONS**

Revision: **1.2**

Date: **20/06/2022**

1 INTRODUCTION

A beam is a structural element that primarily resists loads applied laterally to the beam's axis. Its mode of deflection is primarily by bending. The loads applied to the beam result in reaction forces at the beam's support points. The total effect of all the forces acting on the beam is to produce shear forces and bending moments within the beam, that in turn induce internal stresses, strains and deflections of the beam.

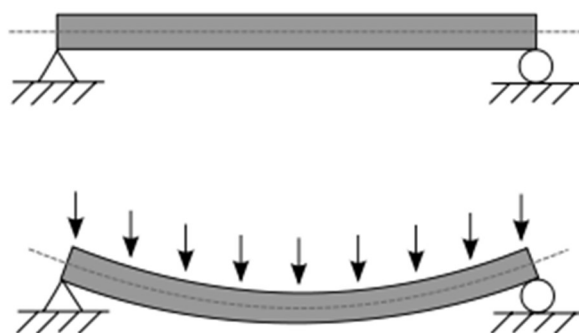


Figure 1. Beam in bending

The beam is long in proportion to the dimensions of the cross-section. Typically this ratio should be at least 10 (*). This chapter will not deal with the calculation of the internal loads (forces and moments) at beam sections, what is left for Volume II of this Stress Methods Manual (see also practical data in Ref. 1 and classical literature), but with the calculation of the stresses within the section, provided the internal loads are known. Methods presented in this chapter shall cover stress calculation of the beam section in a more general sense, i.e. not only subjected to bending moments and shear forces, but also to axial forces and torsion moments. Many paragraphs of this chapter are literal extracts from that classical literature (see list of references).

(*) The slenderness ratio of a uniform beam/column is defined as the ratio of length to the minimum radius of gyration of the cross-section (see definition of radius of gyration later).

1.1 METHOD-SUMMARY

Stresses under Axial Loads and Bending

The primary tool for structural analysis of slender beams is the Euler–Bernoulli beam equation.

Euler–Bernoulli beam theory (also known as engineer's beam theory or classical beam theory) is a simplification of the linear theory of elasticity which provides a means of calculating the load-carrying and deflection characteristics of beams. It covers the case for small deflections of a beam that are subjected to lateral loads only.

Literature suggests, as alternative methodology for calculating shorter beams or cases with high deflections, the Timoshenko beam theory. However, the simpler Euler-Bernoulli theory is applicable to most cases, and for the rest, normally a numerical method (FEM) shall be the most suitable solution. Thus, the Timoshenko beam theory shall not be discussed here.

Shear and Torsion Stresses

Methodologies for calculation shear and torsion stresses in beams are also available in literature in case the beam is made of thin-walled sections, i.e. of sections built on segments with thickness significantly lower than width.

Methods for these configurations are presented in following chapter.

1.2 SCOPE AND LIMITATIONS

Stress analysis of straight beams of isotropic materials.

Out of the scope of methodologies presented in this chapter:

- ✓ Short beams or high deflections
- ✓ Curved beams (i.e. ratio curvature radius to section depth/height lower than 10)
- ✓ Composite beams (i.e. beams made of composite materials)
- ✓ Calculation of internal loads (forces and moments) along the beam sections
- ✓ Beams with “hard” section variations along its span
- ✓ Shear and torsion stress analysis for non-thin-walled-sections

Dedicated methodologies for covering curved beams analysis shall be treated in chapter §3411 “STRESS IN CURVED BEAMS SECTIONS”, and for composite beams in chapter §3420 “COMPOSITE BEAMS SECTIONS”.

2 METHOD DESCRIPTION

2.1 Stresses under Axial Loads and Bending

As noted previously, the Euler–Bernoulli beam theory shall be used to calculate the stress distribution on a section of a beam. The basic principle of this methodology is that cross-sections normal to the neutral axis remain plane and normal to the neutral axis (points in sections with null stress/strain in pure bending).

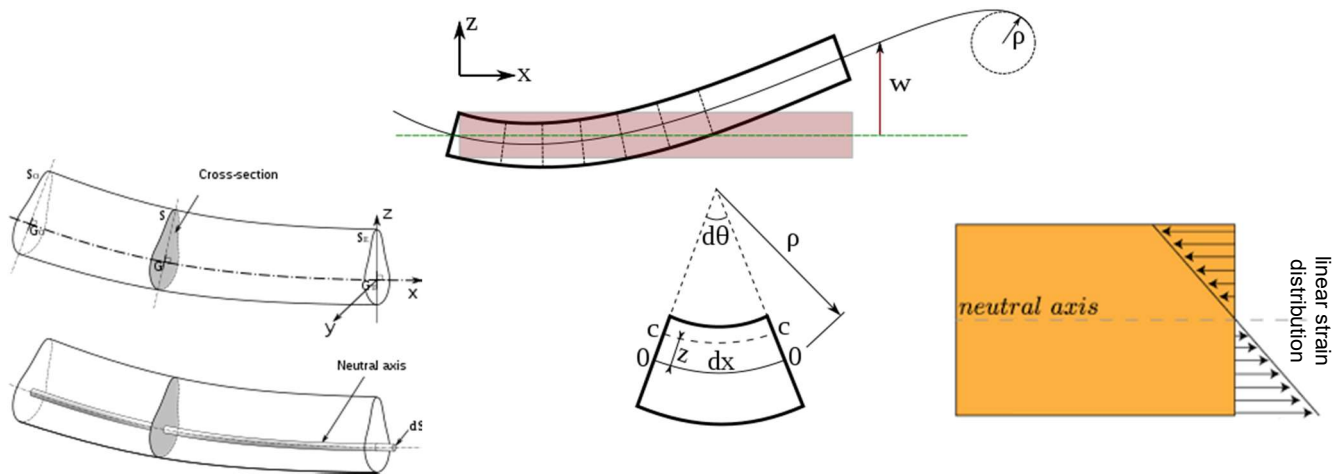


Figure 2. Bending of an Euler–Bernoulli beam

Equilibrium equations

At each section, if X is aligned with the beam axis (axial direction), Y/Z represent the local reference directions within the cross section, the relationships between axial load P_x , and bending moments M_y , M_z , with the axial stress distribution and cross-section geometry can be expressed through integrals across the section of area A as:

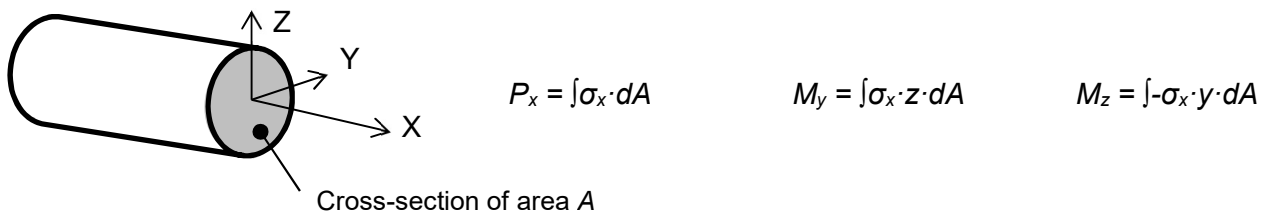


Figure 3. Loads equilibrium in a beam section

Tension and bending in linear elastic isotropic materials

For linear elastic materials the neutral axis passes through the centre of gravity of the sections. Using a local reference axis with origin in the centre of gravity of the section, and with axes matching the

principal axis system (immediate in sections with any axis of symmetry), the axial stress distribution across the section becomes:

$$\sigma_x = P_x / A + M_y \cdot z / I_y - M_z \cdot y / I_z$$

$$\dots \text{where} \quad A = \int dA \quad , \quad I_y = \int z^2 \cdot dA \quad , \quad I_z = \int y^2 \cdot dA$$

Section properties I_y and I_z are its bending moments of inertia (or second moments of area). Tabulated solutions of these values for many typical sections can be found in classical literature, e.g. in Roark's book, Ref. 1, appendix A. Expressions to obtain the values of the principal axis of inertia and their position can be also found in that reference. Additionally, a frequently practical tool needed in analyses is the parallel axis theorem, also known as Huygens–Steiner theorem, or just as Steiner's theorem, which can be used to determine the moment of inertia or the second moment of area about any axis I , given the moment of inertia about a parallel axis through the centre of gravity I_{cg} and the perpendicular distance between the axes d .

$$I = I_{cg} + A \cdot d^2$$

Bending inertia is sometimes defined by the radius of gyration of the cross-section, referred typically as r or ρ in literature. Its relationship with the bending moment of inertia I and the area A is:

$$\rho = \sqrt{I/A}$$

As noted, both strain and stress have a linear response vs applied loads in this case (isotropic linear elastic materials).

If the cross-section has no symmetries in the local reference axis system, the solution for the elastic linear axial stress in combined axial and bending loads can be expressed as (see Bruhn A.13.5):

$$\sigma_x = \frac{P_x}{A} + \frac{M_y I_z + M_z I_{yz}}{I_y I_z - I_{yz}^2} z - \frac{M_z I_y + M_y I_{yz}}{I_y I_z - I_{yz}^2} y$$

...where the third moment of inertia $I_{yz} = \int y \cdot z \cdot dA$

The equation above can be expressed in terms of the *effective bending moments* in the more familiar form as:

$$\sigma_x = \frac{P_x}{A} + \frac{\bar{M}_y}{I_y} z - \frac{\bar{M}_z}{I_z} y$$

...where the *effective bending moments* are defined as:

$$\bar{M}_y = \frac{M_y + M_z I_{yz} / I_z}{1 - \frac{I_{yz}^2}{I_y I_z}} \quad \bar{M}_z = \frac{M_z + M_y I_{yz} / I_y}{1 - \frac{I_{yz}^2}{I_y I_z}}$$

Note the bending/curvature deformations of the section will be then:

$$\theta_{y,x} = \bar{M}_y / (EI_y) \quad \theta_{z,x} = \bar{M}_z / (EI_z)$$

Further discussion and explanations on this topic can be easily found in classical literature.

Tension and bending in inelastic isotropic materials

If the beam material is not perfectly linear elastic, which is typical of metallic beams, but inelastic, the stress distribution will not be linear across the section even the Euler–Bernoulli is applied (and therefore strain response remains linear).

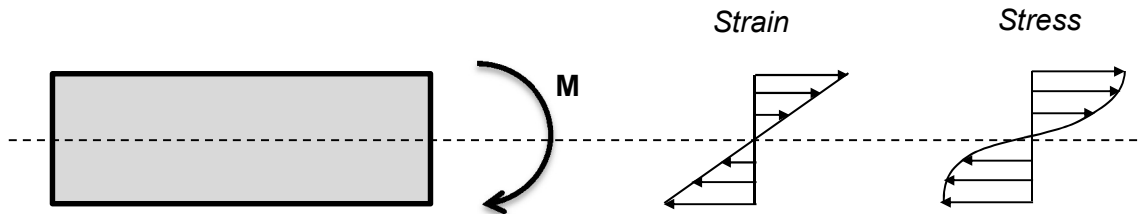


Figure 4. Bending strain and stress in a beam made of inelastic isotropic material

The ultimate bending moment of a section is a function of the cross-sectional shape. The shape can be represented by the factor k which is obtained by the following formula:

$$k = 2 \cdot Q / (I/c)$$

...where:

- ✓ Q is the static moment, or the first moment of area, from the section most upper fibre to the principal axis if bending is around the local Y principal axis, and it is obtained as:

$$Q = \int_{z_{cg}}^{z_{max}} z \cdot dA$$

- ✓ I is the bending moment of inertia of the cross-section relative to its principal axis
- ✓ c is the distance from the principal axis to the extreme fibre

k	1.0	1.5	1.0 - 1.5	1.0 - 1.5	1.27 - 1.7	1.5 - 2.0①
Section						

Figure 5. Range of values of section factors k

Using this section factors (or shape factors) and approaches as the *Cozzone's* method (Bruhn §C3.4) or graphs in literature (Niu §6.2), the ultimate strength of some typical beam sections in bending can be calculated.

However, in a general case no explicit solution shall be available for the stress, and a numeric method as that proposed in ESDU90004 shall have to be applied.

2.2 Shear and Torsion Stresses in thin-walled sections

Shear stress

Shear stresses in thin-walled sections can be calculated thanks to the classical theory of elasticity and Collignon's theorem (shear stresses are uniform across the thickness).

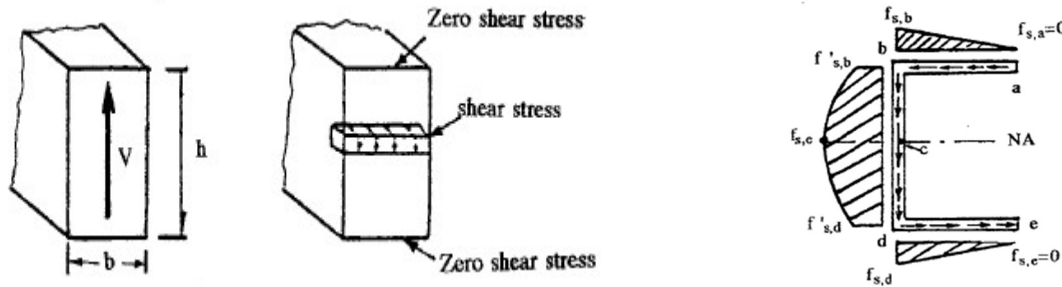


Figure 6. Shear load and stress in a simple rectangular section and in a C-shape section

The formula to obtain the stress distribution in a segment of thin-walled section subjected to a (vertical) shear force is:

$$\tau = V_z \cdot Q_y / (I_y \cdot t)$$

...where:

- ✓ V is the applied shear force
- ✓ Q is the static moment, or the first moment of area, from the section most upper fibre to the point of interest, obtained as:

$$Q_y = \int_z^{z_{max}} z \cdot dA$$

- ✓ I is the bending moment of inertia of the cross-section relative to its neutral axis (CG)
- ✓ t is the thickness of the wall segment

For thin-walled closed sections there are not free edges and thus, not necessarily section points with 0 shear flow/stress. The standard technique to solve the shear problem in this kind of section is to split the section at a segment, add a constant shear flow as unknown, superimpose those stress distributions, and state the forces equilibrium (shear force and torsion moment) to find the solution. The approach for multicellular sections is similar. Check classical literature for further details and examples of calculations.

Shear centre

Bending without twisting is possible, provided that the loads are in the same plane that the resultant shear stresses act, or in more general terms, bending moment without twisting will occur when the loads are applied through the shear centre. The shear centre is defined as the point of the cross section of a member through which the resultant of the shear stresses must pass for any orientation of loading so that the member will bend but not twist.

Torsion stresses

For uniform straight members in pure torque for unrestrained torsion in the elastic range, the applied torque is balanced by the section itself. Twisting of any section along the member is about a longitudinal axis of rotation passing through the shear centre. The rate of change of the angle of twist can be obtained as:

$$\theta / L = T / (G \cdot J)$$

...where:

- ✓ θ is the angle of twist
- ✓ L is the length of the beam
- ✓ T is the applied torsional moment
- ✓ G is the shear modulus of the material
- ✓ J is the torsional constant/stiffness, dependent on the form of the cross section (for a circular section equal to the polar moment of inertia)

Thin-walled closed sections

Torsion of thin-walled closed section members is balanced at any section by a uniform shear flow around the median line of the section. The stress is assumed to be constant across the thickness at any point on the section.

The torsional constant is given by:

$$J = 4 \cdot A^2 / \Sigma(b/t_i)$$

...where:

- ✓ A is the area enclosed by the median contour
- ✓ b_i is the width of the wall segment
- ✓ t_i is the thickness of the wall segment

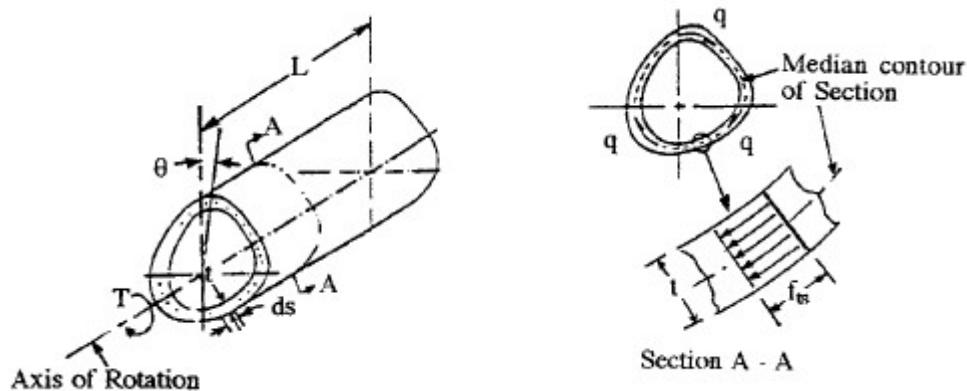


Figure 7. Torsion of thin-walled closed sections

And the torsion stress is obtained as:

$$\tau = T / (2 \cdot A \cdot t)$$

Thin-walled open sections

In a thin-walled section subjected to torsion the stress distribution is assumed to be linear across the section.

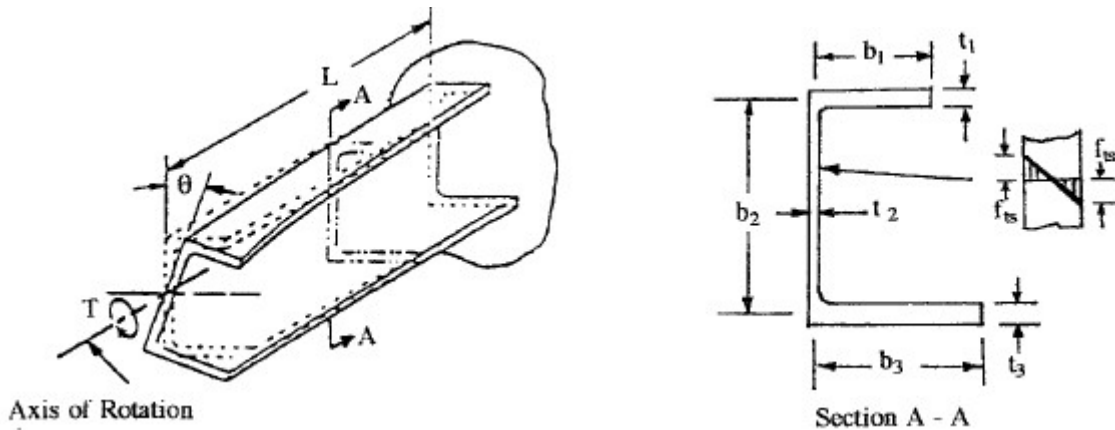


Figure 8. Torsion of thin-walled open sections

The torsional constant is given by:

$$J = \sum (b_i \cdot t_i^3 / 3)$$

And the torsion stress is obtained as:

$$\tau = T \cdot t / J$$

Other considerations about torsion

Effect of end constraint

Extract from Roark's §10.3: "If one or both ends of a bar are so fixed that warping is prevented, or if the torque is applied to a section other than at the ends of a bar, the stresses and the angle of twist produced by the given torque are affected. In compact sections the effect is slight, but in the case of open thin-walled sections the effect may be considerable".

That reference provides extended formulation for taking into consideration this effect in a limited set of boundary conditions and applied torsion loads, on thin-walled sections of straight beams, including expressions for evaluating the warping stiffness factor C_w , the so-called β function (equal to $[GJ/C_w E]^{1/2}$), and the resulting corrected twist angles.

Effect of Longitudinal Stresses

The elongation of the outer fibers consequent upon twist will cause also longitudinal stresses. The effect of induced longitudinal stress will be more relevant in any bar having a section composed of narrow rectangles, such as a I, T, or channel. This effect may stiffen and strengthen the beam in torsion, but also add longitudinal compression stresses, thus made the section flanges more prone to buckle.

This effect will not be discussed here.

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
A	mm^2	Area of the cross-section, or area enclosed by the median contour in a thin-walled closed section, from context
b	mm	Width of a thin-walled section segment
c	mm	Distance from the principal axis to the extreme fibre
CG, cg	-	As subscript, centre of gravity
C_w	mm^6	Warping stiffness factor
d	mm	Distance between two parallel axis
E	MPa	Young modulus of the material
G	MPa	Shear modulus of the material
I	mm^4	Bending moment of inertia, or second moment of area, of the cross-section
J	mm^4	Torsional constant/stiffness of the cross-section
k	-	Section factor
M	N/mm	Bending moment
P_x	N	Axial force
Q	mm^3	Static moment, or first moment of area, of the cross-section
r, ρ	mm	Radius of gyration of the section
t	mm	Thickness of a thin-walled section segment
T	Nmm	Torsion moment
TBC		To Be Confirmed
V	N	Shear force
X, Y, Z	-	Directions of a rectangular reference axis system (also used as subscripts)

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[W. C. Young & R. G. Budynas]	7 th Ed.	ROARK'S FORMULAS FOR STRESS AND STRAIN
Ref. 2	ESDU90004	1990	MAXIMUM INELASTIC BENDING MOMENTS AND END LOADS (IN FIXED RATIO) FOR BEAM SECTIONS OF GENERAL POLYGONAL SHAPE
Ref. 3	[M. C. Y. Niu]	2 nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 4	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES

SOFTWARE

Program	Version	Description
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures (Modules BEAM & ESDU90004)
FLEXA (Airbus DS)	3.9	Strength check for constant section beams subjected to axial load and bending in plastic regime

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
1.0 01/08/2018	G. Baños	First issue	All pages
1.1 10/09/2018	G. Baños	Fix error in expression for the torsional constant of thin-walled open sections	Page 7
1.2 20/06/2022	G. Baños	Add explicit formulation for linear bending of non-symmetric sections	Page 4

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Methods (TEPAP-TL4)	G. Baños 20/06/2022	F. Glock 26/06/2022

Table 3. *Approval signatures table for last revision of this chapter*